

An Explicit Bijection for the k -Block Index

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ABSTRACT. For an integer $k \geq 2$, Wang and Zhang defined a k -block index on k -regular partitions and stated an identity equating its joint distribution with the length to the joint distribution of the k -alternating length and the length on k -strict partitions. A direct bijection is given. The map iterates a simultaneous transfer which lowers the largest part of every nonterminal block by k and deletes one part from the terminal block. The successive block indices form a partition whose adjacent differences are less than k . Its conjugate is k -strict. An explicit reverse block transfer establishes bijectivity and gives the required preservation of size and both statistics.

1. INTRODUCTION

For a partition λ , write $|\lambda|$ for its size and $\ell(\lambda)$ for its length. Wang and Zhang introduced the k -block index for k -regular partitions in their study of refinements of Euler's partition theorem [2]. Their definition extends the block-index setting considered in [1]. In Section 7.2 of [2], they stated the following identity and left its proof to the reader:

$$\sum_{\lambda \in \mathcal{R}_k} x^{\ell(\lambda)} y^{b_k(\lambda)} q^{|\lambda|} = \sum_{\pi \in \mathcal{S}_k} x^{\ell_k(\pi)} y^{\ell(\pi)} q^{|\pi|}. \quad (1.1)$$

Here $\mathcal{R}_k$, $\mathcal{S}_k$, b_k , and ℓ_k are defined in Section 2. The purpose of this paper is to give a direct bijective proof of (1.1).

The construction is local. A simultaneous transfer acts on every block of a k -regular partition. Repeating the transfer records a decreasing sequence of block indices. The central point is that consecutive entries differ by at most $k - 1$. Consequently, conjugation of the sequence produces a k -strict partition. The inverse is obtained by a reverse transfer that groups parts from the bottom upward. The proof also identifies the two statistics without recourse to a product expansion.

2. DEFINITIONS AND STATEMENT OF THE RESULT

A partition is a finite weakly decreasing sequence of positive integers. Its parts are understood with multiplicity. Fix an integer $k \geq 2$. A partition is k -regular if none of its parts is divisible by k , and it is k -strict if every part occurs fewer than k times. Let $\mathcal{R}_k$ and $\mathcal{S}_k$ denote the corresponding sets, including the empty partition.

Let $\lambda \in \mathcal{R}_k$ be nonempty. Its k -blocks are obtained by working from the largest unused part downward. If a is the largest part not yet assigned, the next block contains all remaining parts with values in the interval $[a - k, a]$. The block has weight k when $a > k$, and weight a when $a \leq k$. The k -block index is the sum of the block weights and is denoted by $b_k(\lambda)$; set $b_k(\emptyset) = 0$. Since the only possible block with largest part at most k is the final block, there is at most one block whose weight is less than k .

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For $\pi = (\pi_1, \pi_2, \dots) \in \mathcal{S}_k$, extend the sequence by $\pi_j = 0$ past its length and define

$$\ell_k(\pi) = \sum_{j \geq 0} (\pi_{jk+1} - \pi_{(j+1)k}). \quad (2.1)$$

The sum is finite. This is the statistic used in [2].

Theorem 2.1. *For every integer $k \geq 2$, there is a bijection $\Phi_k : \mathcal{R}_k \rightarrow \mathcal{S}_k$ such that, for every $\lambda \in \mathcal{R}_k$,*

$$|\Phi_k(\lambda)| = |\lambda|, \quad \ell(\Phi_k(\lambda)) = b_k(\lambda), \quad \ell_k(\Phi_k(\lambda)) = \ell(\lambda). \quad (2.2)$$

Consequently, (1.1) holds.

3. THE BLOCK TRANSFER

For a partition ν , let

$$\text{supp}(\nu) = \{a \geq 1 : m_a(\nu) > 0\},$$

where $m_a(\nu)$ is the multiplicity of the part a . The transfer below changes multiplicities, while the block decompositions are determined by this finite support set.

Lemma 3.1 (Greedy interval lemma). *Let D be a finite set of integers and let $k \geq 1$. Partition D by the following two greedy procedures:*

- (1) *start with the least unused element b , take all unused elements in $[b, b+k]$, and continue upward;*
- (2) *start with the greatest unused element a , take all unused elements in $[a-k, a]$, and continue downward.*

The two procedures produce the same number of blocks.

Proof. In the first procedure, record the least element selected at each step. These recorded elements differ pairwise by more than k , and every subset of D whose distinct elements differ pairwise by more than k contains at most one element in each greedy block. Thus the number of blocks in the first procedure is the maximum cardinality of a k -separated subset of D . The greatest-first procedure has the same property, using its greatest elements as the recorded separated subset. Hence the two block counts agree. $\square$

Let $\lambda \in \mathcal{R}_k$ be nonempty. Define $\tau(\lambda)$ by acting simultaneously on its k -blocks. From every block with largest part $a > k$, replace one occurrence of a by $a - k$. From the final block with largest part $a \leq k$, if it exists, delete one occurrence of a . The resulting partition is again k -regular. The decrease in size is the sum of the block weights:

$$|\tau(\lambda)| = |\lambda| - b_k(\lambda). \quad (3.1)$$

The reverse operation is explicit. For an integer $w > 0$, let r be its least nonnegative residue modulo k .

Lemma 3.2 (Explicit reverse transfer). *Let $\mu \in \mathcal{R}_k$, put $h = b_k(\mu)$, and suppose that*

$$h \leq w \leq h + k - 1. \quad (3.2)$$

Construct $\sigma_w(\mu)$ by the following finite algorithm.

- (1) *If $r > 0$, adjoin one part r to μ .*
- (2) *Leave all parts of size at most r fixed. On the distinct values in $\text{supp}(\mu)$ larger than r , work from the smallest upward. If b is the smallest unprocessed value, make a reverse block from all unprocessed values in $[b, b+k]$, and continue with values larger than $b+k$.*
- (3) *In every reverse block, replace one occurrence of its smallest part b by $b+k$.*

Then $\sigma_w(\mu) \in \mathcal{R}_k$ and

$$\tau(\sigma_w(\mu)) = \mu, \quad b_k(\sigma_w(\mu)) = w. \quad (3.3)$$

Proof. The algorithm is deterministic and terminates because the support is finite. Every new value has the same nonzero residue modulo k as the value it replaces, so $\sigma_w(\mu)$ is k -regular.

Write

$$h = ak + t, \quad 0 \leq t < k. \quad (3.4)$$

There are a blocks of μ whose largest parts exceed k ; if $t > 0$, the remaining block has largest part t . Let

$$D_r = \{d \in \text{supp}(\mu) : d > r\}.$$

Removing the values at most r removes the terminal block when $r \geq t$, and leaves a nonempty part of it when $r < t$. The blocks with largest parts greater than k retain their largest parts, so the greatest-first decomposition of D_r has a blocks when $r \geq t$, and $a + 1$ blocks when $r < t$. By Lemma 3.1, the reverse-block algorithm in step 2 has the same number, say q , of blocks. Consequently $q = a$ in the first case and $q = a + 1$ in the second. The interval condition (3.2) gives $w = ak + r$ in the first case and $w = (a + 1)k + r$ in the second. Therefore $qk + r = w$.

Let the reverse blocks have smallest values $b_1 < \dots < b_q$. Consecutive smallest values satisfy $b_{j+1} > b_j + k$. After step 3, the values originating in the j th reverse block lie in $[b_j, b_j + k]$ and include $b_j + k$. Moreover, $b_j + k < b_{j+1}$, so the top-down decomposition cannot merge two adjacent reverse blocks. It therefore recovers these intervals, in reverse order, as blocks with largest values $b_q + k, \dots, b_1 + k$. If $r > 0$, all fixed values are at most r and the adjoined value r is present. When $q > 0$, they lie below b_1 and form the final block; when $q = 0$, they form the only block. In both cases its largest value is r .

Thus the block weights of $\sigma_w(\mu)$ are q copies of k and, when $r > 0$, one weight r ; their sum is $qk + r = w$. Applying τ lowers one $b_j + k$ to b_j in every nonterminal block and removes the adjoined part r from the terminal block. All multiplicities are thereby restored, so $\tau(\sigma_w(\mu)) = \mu$. $\square$

Lemma 3.3 (Forward transfer is inverted by the algorithm). *Let $\lambda \in \mathcal{R}_k$ be nonempty, put $w = b_k(\lambda)$, and set $\mu = \tau(\lambda)$. Then*

$$b_k(\mu) \leq w \leq b_k(\mu) + k - 1, \quad (3.5)$$

and $\lambda = \sigma_w(\mu)$.

Proof. Write $w = ak + r$ with $0 \leq r < k$. The a nonterminal blocks of λ have largest parts $A_1 > \dots > A_a > k$. If $r > 0$, the final block has largest part r ; if $r = 0$, there is no final block of weight less than k .

For $1 \leq j \leq a$, put $c_j = A_j - k$. The image under τ of the j th nonterminal block has support contained in $[c_j, c_j + k]$ and contains c_j , because one occurrence of A_j was moved to $A_j - k$. For $1 \leq j < a$, the next lower nonterminal block satisfies

$$c_{j+1} + k = A_{j+1} < A_j - k = c_j,$$

by the defining greedy separation of the original blocks. Thus the support intervals $[c_j, c_j + k]$ are pairwise separated. If a terminal block exists, its largest part is $r < c_a$; after the deletion, every surviving terminal value is at most r .

Consequently, the distinct values of μ larger than r are partitioned by the reverse greedy procedure into exactly the a transformed nonterminal blocks. It starts at c_a , recovers the support in $[c_a, c_a + k]$, and then proceeds successively through the separated intervals up to $[c_1, c_1 + k]$. The values at most r are precisely the unchanged terminal values. Step 3 of the construction of $\sigma_w(\mu)$ therefore reverses every move made by τ , including the possible terminal deletion. Hence $\lambda = \sigma_w(\mu)$.

It remains to compare block indices. Set

$$D_r = \{d \in \text{supp}(\mu) : d > r\}.$$

If D_r is empty, then $a = 0$ and all parts of μ are at most r . Hence $b_k(\mu) = t$ for some $0 \leq t \leq r$, where $t = 0$ means that μ is empty.

Suppose that D_r is nonempty. Its bottom-up decomposition has the a transformed nonterminal blocks identified above, so [Lemma 3.1](#) shows that its top-down decomposition also has exactly a blocks. Since μ is k -regular, the top of its lowest block cannot equal k . If that top is greater than k , then all a blocks have weight k . Values at most r either join the lowest block or form a terminal block of top $t \leq r$; in either event

$$b_k(\mu) = ak + t \quad (0 \leq t \leq r),$$

with $t = 0$ when there is no terminal block. If the lowest block has top less than k , it is itself the terminal block and contains every positive value of μ at most r . Its top t satisfies $r < t < k$, while the other $a - 1$ blocks have weight k . Thus

$$b_k(\mu) = (a - 1)k + t \quad (r < t < k).$$

The first formula also covers $D_r = \emptyset$. In the two cases, respectively,

$$w - b_k(\mu) = r - t, \quad w - b_k(\mu) = k + r - t.$$

Both differences lie in $\{0, 1, \dots, k - 1\}$, which proves [\(3.5\)](#). $\square$

4. THE BIJECTION AND ITS STATISTICS

Start with $\lambda^{(0)} = \lambda \in \mathcal{R}_k$ and define

$$\lambda^{(i+1)} = \tau(\lambda^{(i)}) \quad \text{while } \lambda^{(i)} \neq \emptyset. \quad (4.1)$$

The process terminates by [\(3.1\)](#). Let

$$\mathbf{s}(\lambda) = (s_1, s_2, \dots, s_m), \quad s_i = b_k(\lambda^{(i-1)}), \quad (4.2)$$

where $\lambda^{(m)} = \emptyset$. Define

$$\Phi_k(\lambda) = \mathbf{s}(\lambda)'. \quad (4.3)$$

Proposition 4.1. *Let $\lambda \in \mathcal{R}_k$ be nonempty. The sequence $\mathbf{s}(\lambda)$ is a partition and satisfies*

$$0 \leq s_i - s_{i+1} < k \quad (1 \leq i < m), \quad 0 < s_m < k. \quad (4.4)$$

Moreover, $\Phi_k(\lambda)$ is k -strict and satisfies the three identities in [\(2.2\)](#).

Proof. Apply [Lemma 3.3](#) at every step of [\(4.1\)](#). It gives the first inequality in [\(4.4\)](#). In the final nonempty partition all remaining parts lie in a single block with top less than k , so $0 < s_m < k$.

For a partition $\mathbf{s}$, the multiplicity of the part i in its conjugate is $s_i - s_{i+1}$, where $s_{m+1} = 0$. Thus [\(4.4\)](#) implies that $\mathbf{s}'$ is k -strict. Telescoping [\(3.1\)](#) over the transfer sequence gives

$$|\lambda| = s_1 + s_2 + \dots + s_m = |\mathbf{s}'|. \quad (4.5)$$

The length of $\mathbf{s}'$ is its largest part $s_1 = b_k(\lambda)$.

It remains to identify the length statistic. A transfer changes no number of parts except when a terminal block is present, in which case it deletes one part. Such a block is present exactly when the corresponding s_i is not a multiple of k . Hence

$$\ell(\lambda) = \#\{i : k \nmid s_i\}. \quad (4.6)$$

Let $\pi = \mathbf{s}'$. For every $j \geq 0$,

$$\pi_{jk+1} - \pi_{(j+1)k} = \#\{i : jk < s_i < (j+1)k\}. \quad (4.7)$$

Summing [\(4.7\)](#) and using [\(2.1\)](#) shows that its left-hand side is $\ell_k(\pi)$ and its right-hand side is the quantity in [\(4.6\)](#). This proves all three identities. $\square$

Proof of Theorem 2.1. The empty partition maps to itself. For a nonempty λ , the preceding proposition shows that Φ_k has the required statistics. It remains to prove that the map is a bijection on nonempty partitions. Let $\pi \in \mathcal{S}_k$ be nonempty and put $\mathbf{s} = \pi' = (s_1, \dots, s_m)$. Since π is k -strict,

$$0 \leq s_i - s_{i+1} < k \quad (1 \leq i < m), \quad 0 < s_m < k. \quad (4.8)$$

Set $\mu^{(m)} = \emptyset$. For $i = m, m-1, \dots, 1$, define recursively

$$\mu^{(i-1)} = \sigma_{s_i}(\mu^{(i)}). \quad (4.9)$$

Inductively, $b_k(\mu^{(i)}) = s_{i+1}$, with $s_{m+1} = 0$. Thus (4.8) is exactly the admissibility condition in Lemma 3.2. That lemma gives $\tau(\mu^{(i-1)}) = \mu^{(i)}$ and $b_k(\mu^{(i-1)}) = s_i$. Therefore $\mathbf{s}(\mu^{(0)}) = \mathbf{s}$, so $\Phi_k(\mu^{(0)}) = \pi$. The same reverse recursion is forced by Lemma 3.3; hence the preimage is unique. This proves bijectivity and the theorem. $\square$

5. AN EXAMPLE

Take $k = 4$ and $\lambda = (10, 7, 3, 1)$. Its first block is $(10, 7)$ and its terminal block is $(3, 1)$, so $b_k(\lambda) = 7$. Iterating the transfer gives the following table.

TABLE 1. The transfer sequence for $k = 4$ and $\lambda = (10, 7, 3, 1)$.

i	$\lambda^{(i)}$	$b_k(\lambda^{(i)})$
0	$(10, 7, 3, 1)$	7
1	$(7, 6, 1)$	5
2	$(6, 3)$	4
3	$(3, 2)$	3
4	(2)	2

Thus $\mathbf{s}(\lambda) = (7, 5, 4, 3, 2)$ and

$$\Phi_4(\lambda) = (5, 5, 4, 3, 2, 1, 1).$$

This partition is 4-strict. Its size is 21, its length is 7, and

$$\ell_4(\Phi_4(\lambda)) = (5 - 3) + (2 - 0) = 4 = \ell(\lambda),$$

which displays the three statistic equalities in Theorem 2.1.

REFERENCES

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